

1. Desktop Client (The "Auditor App")

- **UI Framework: Avalonia UI** (FOSS). It is the modern, hardware-accelerated successor to WPF, delivering high-performance grids that stay smooth even with 100k+ rows of audit data.
- **Architecture Pattern: CommunityToolkit.Mvvm**. This provides a clean separation of your UI (View) and your Audit logic (ViewModel), making the app easier to test and maintain.
- **Language: C# 14** on **.NET 10 (LTS)**. You get the latest language features like the field keyword for cleaner data models and long-term security support from Microsoft.
- **Local Database: SQLite** with **Open-Source SQLCipher**. This keeps your local data encrypted using 256-bit AES without the \$999 commercial license fee.

2. Data & Synchronization

- **Sync Logic: Dotmim.Sync**. A "blazing fast" framework that handles the bi-directional sync between the auditor's laptop and the central server, including conflict resolution if two people edit the same file offline.
- **Central Database: PostgreSQL**.
 - **Phase 1:** Hosted on **Supabase (Free Tier)** for LKR 0.
 - **Phase 2:** Migrated to **Azure Database for PostgreSQL** for advanced enterprise security when the firm scales.
- **Object Storage: Cloudflare R2**. Used for storing all audit evidence, scanned PDFs, and registration documents. It has **Zero Egress Fees**, meaning you aren't charged when auditors download large files.

3. Backend & Infrastructure

- **Server Logic: ASP.NET Core Web API**. The industry standard for high-throughput, secure enterprise backends.
- **Hosting: Hetzner Cloud (Germany/Finland)**.
 - **Plan:** CX21 (2 vCPU, 4GB RAM).
 - **Benefit:** 5–10x cheaper than Azure/AWS with strict ISO 27001 German security compliance.
- **SMS Integration: Notify.lk (Lite Plan)**. For automated client notifications during company registration stages.

4. Deployment & Updates

- **Installer & Updater: Velopack.**
 - It provides a "one-click" installer and handles **silent background updates**.
 - It uses **Small Delta Updates**, meaning if you fix a tiny bug, the auditor only downloads a 1MB "diff" instead of a fresh 100MB installer.